

Paradigm Persistence as Two-Timescale Active Inference

A categorical POMDP with a fast world state and a slow,
theory-laden paradigm belief

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Abstract

Scientific communities revise theoretical commitments more slowly than the evidence alone would dictate. We model this lag with a categorical partially observed Markov decision process built on two latent variables operating at separate timescales. A fast world state s_t , with nontrivial dynamics, is the regime actually realised in an experiment; a slow paradigm θ , with a near-identity transition, indexes the likelihood through which observations are read, so that the same world state yields different outcome distributions under different paradigms. Theory-ladenness is exactly this dependence of the likelihood on θ , and the truth of a paradigm reduces to its predictive goodness-of-fit. Inference proceeds at both timescales: the agent filters the world state by ordinary, untempered Bayes within each paradigm, and revises its paradigm belief by a slow, motivated variational update that trades the accuracy of the marginal likelihood and an affective utility against a conservatism cost. That cost—the divergence from the prior to the posterior over θ —is the informational work of re-reading the accumulated record under a new paradigm. We embed the agent in a network, following the epistemic-communities account, in which agents gain epistemic value from sampling like-minded neighbours, and we take the population distribution of paradigm beliefs as the macroscopic order parameter. With one-round transmission delays the population is a second-order system whose approach to consensus is a damped oscillation governed by the distribution of conservatism and the network structure; permanent lock-in is a singular extreme. We specify the model, visualise the trust network from the graph construction used in the implementation, and present illustrative schematics of the target dynamics for a forthcoming numerical study.

1 Introduction

The persistence of scientific paradigms beyond the point at which evidence begins to favour an alternative is a well-documented feature of the history of science (Kuhn, 1962; Planck, 1949). A purely Bayesian agent revises beliefs in proportion to the strength of evidence; the observed lag therefore indicates that belief revision carries costs the cost-free Bayesian calculus omits (Hyland and Albarracin, 2025).

The model rests on a separation of two hidden quantities at distinct timescales. The fast hidden state s_t is the regime realised in an experiment and carries nontrivial dynamics; the slow latent θ is the paradigm, which the agent does not sample but maintains a belief over, and which enters the model as the likelihood through which the world state is read. A paradigm is thus not a value of the world but a mapping from world states to observation distributions.

This separation underlies the account that follows. Inference over the world state is untempered and fast, whereas revision of the paradigm is slow and costly, because changing the belief over θ alters the interpretation of every datum already recorded. In the variational account this retroactive cost is the complexity term of the free energy, the divergence from the prior to the posterior over θ ; following Hyland and Albarracin (2025) we identify it with the cost of changing one’s mind. We further place the agent in a network of peers, after Albarracin et al. (2022), so that paradigm beliefs propagate, and we study the population distribution of those beliefs.

Section 2 reviews the components. Section 3 specifies the two-timescale generative model, the two-level inference scheme, the trust coupling (with a visualisation of the network and its

matrices from the graph construction used in the implementation), and the regimes of slow paradigm change the model is built to exhibit. Section 4 presents illustrative schematics of the target dynamics. Section 5 states the layered explanation of persistence the model affords and the experimental programme it implies.

2 Background

Variational free energy. An active-inference agent maintains an approximate posterior q over hidden states and minimises the variational free energy of an observation o ,

$$F[q, o] = \underbrace{-\mathbb{E}[\log p(o | s)]_q}_{\text{accuracy}} + \underbrace{D_{\text{KL}}[q(s) \| p(s)]}_{\text{complexity}}, \quad (1)$$

the sum of an accuracy term and a complexity term measuring divergence of the posterior from the prior (Friston et al., 2015). Action is selected to minimise expected free energy.

Motivated belief revision. Hyland and Albarracin (2025) reinterpret the complexity term of Eq. (1) as a substantive cost of changing one’s mind, and recast belief updating as the maximisation of a belief-utility net of that cost. The belief-utility decomposes into an affective component, independent of accuracy, and an accuracy component; a single conservatism parameter scales the cost, ranging from costless revision to immovable belief. The framework reproduces confirmation bias and attitude polarisation.

Epistemic communities. Albarracin et al. (2022) model active-inference agents on a graph who gain epistemic value from observing their neighbours’ communicated beliefs; an epistemic confirmation bias makes agents value like-minded neighbours more, driving the formation of belief clusters.

3 Model

3.1 Two latents on two timescales

The generative model has two hidden factors. The *world state* $s_t \in \mathcal{S}$ is the regime realised at time t ; it has nontrivial dynamics $B_s(s_t | s_{t-1}, a_{t-1})$ driven by the agent’s experiment a_{t-1} . The *paradigm* $\theta \in \{0, \dots, K-1\}$ is a slow latent with a near-identity transition $B_\theta \approx I$; it is not resampled but inferred. The model factorises as

$$p(o_{1:T}, s_{1:T}, \theta) = p(\theta) \prod_t p(o_t | s_t, \theta) p(s_t | s_{t-1}, a_{t-1}), \quad (2)$$

and the agent’s posterior factorises accordingly, $q(s_{1:T}, \theta) = q(\theta) \prod_t q(s_t | \theta)$. The separation of timescales is the defining feature of the model: the world state evolves quickly and the paradigm slowly, and the agent infers both. Figure 1 shows the structure.

3.2 Theory-laden likelihood

For each paradigm θ and world state s , the likelihood $p(o | s, \theta)$ is a categorical distribution over a finite outcome set $\mathcal{O}$. Theory-ladenness is the dependence of this distribution on θ : the same world state s produces different outcome distributions read through different paradigms, so no two paradigms interpret a datum identically. We make no metaphysical claim that the world possesses a true paradigm. The world generates observations through some θ^* , which need not lie in the agent’s hypothesis set; the agent maintains a posterior over which θ best predicts its

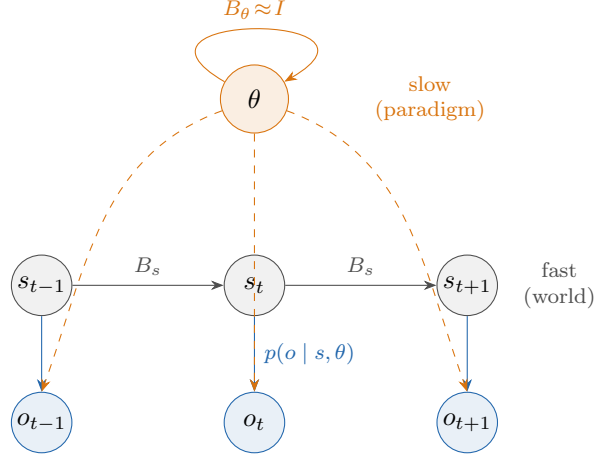


Figure 1: The two-timescale generative model. The world state s_t (grey) evolves under nontrivial dynamics B_s and emits observations through the likelihood $p(o_t | s_t, \theta)$ (blue). The paradigm θ (orange) is a slow latent ($B_\theta \approx I$) that conditions every likelihood (dashed): it does not move the world, it sets how the world is read. The agent infers both factors.

experience, and the question “which paradigm is true” reduces to predictive goodness-of-fit. The capacity of an experiment to discriminate paradigms,

$$d(a) = \frac{1}{2} \sum_{\theta \neq \theta'} D_{\text{KL}} [p(o | s, \theta) \| p(o | s, \theta')], \quad (3)$$

is small when paradigms predict similar outcomes (the experiment is underdetermined) and large when they predict different ones.

3.3 Two-level inference

Both latents are revised by the *same* update—the motivated belief-utility maximisation of Hyland and Albarracin (2025)—and the two timescales are two settings of its parameters, not two mechanisms. For a belief q over a latent with carried-over prior q_0 and evidence summarised by a marginal likelihood L ,

$$q' = \arg \max_q \left\{ U[q] + \alpha \mathbb{E}[\log L]_q - \lambda D_{\text{KL}}[q \| q_0] \right\} \implies q' \propto q_0 L^{\alpha/\lambda} e^{u/\lambda}, \quad (4)$$

a tempered Bayesian rule with effective learning rate α/λ : the accuracy weight α scales the pull of the evidence, the conservatism λ the cost of moving, and the affective utility u (with $U[q] = \mathbb{E}[u]_q$) a preference over which value is believed, independent of fit. Ordinary Bayes is the corner $\alpha = \lambda$, $u = 0$, where the learning rate is one and the prior is updated by the likelihood alone.

Fast: untempered filtering of the world state. The world state sits at that corner. Its prior is the one-step prediction $\sum_{s_{t-1}} B_s(s_t | s_{t-1}, a_{t-1}) q(s_{t-1} | \theta)$ —present because B_s has nontrivial dynamics—and at $\alpha = \lambda$, $u = 0$ Eq. (4) collapses to the exact Bayes filter

$$q(s_t | \theta) \propto p(o_t | s_t, \theta) \sum_{s_{t-1}} B_s(s_t | s_{t-1}, a_{t-1}) q(s_{t-1} | \theta). \quad (5)$$

The world state carries no conservatism because there is no standing commitment to resist revising: the regime is expected to change at every step, so tracking it responsively is correct, not motivated.

Slow: motivated revision of the paradigm. The paradigm sits away from the corner, with $\alpha < \lambda$ and possibly $u(\theta) \neq 0$. Its prior is the carried-over belief $q(\theta)$ —there is no prediction step, since $B_\theta \approx I$ —and the evidence is the marginal likelihood each paradigm assigns the new datum,

$$L(\theta \mid o_t) = \sum_{s_t} p(o_t \mid s_t, \theta) q(s_t \mid \theta). \quad (6)$$

Instantiating Eq. (4) gives the paradigm update

$$q'(\theta) = \arg \max_q \left\{ U[q] + \alpha \mathbb{E}[\log L(\theta \mid o_t)]_q - \lambda D_{\text{KL}}[q \parallel q(\theta)] \right\} \propto q(\theta) L(\theta \mid o_t)^{\alpha/\lambda} e^{u(\theta)/\lambda}. \quad (7)$$

The asymmetry between the factors is therefore a choice of *where* conservatism applies, not a second kind of inference: the world state is tracked responsively because it moves, while the paradigm—a near-static commitment—is revised only as fast as its cost permits.

Conservatism as an institutional-memory cost. What does the conservatism λ encode? As written in Hyland and Albarracin (2025), the cost is the one-step divergence $D_{\text{KL}}[q'(\theta) \parallel q(\theta)]$ between successive paradigm posteriors—the immediate effort of moving one’s belief now. We read its magnitude as a proxy for a larger, retroactive operation. Because θ conditions the likelihood of every past datum (Eq. 2), adopting a new paradigm means re-reading the entire accumulated record under a new likelihood. For a single agent this is the cost of recomputing how all prior observations should now be understood; for a community it is an *institutional-memory* cost—textbooks, curricula, calibrated instruments, and accepted exemplars are encodings of the prevailing θ , and a paradigm shift devalues that shared capital and forces its reconstruction. We are explicit that this archive-recoding reading is an interpretation we place on λ , not a separate term in the objective: Eq. (7) computes only the one-step KL, and we take its size to stand in for the retroactive work, with larger λ marking paradigms whose record is more expensive to recode. So read, λ supplies a formal counterpart of incommensurability—resistance to paradigm change as the informational cost of re-interpreting the past, not a departure from rationality—while we note that a literal accounting of that cost would sum over the history rather than compare two adjacent posteriors.

3.4 Policy and selective attention

Experiments are selected by expected free energy (Friston et al., 2015). The data that move $q(\theta)$ are the discriminating ones—outcomes improbable under every world state for the current paradigm—and these are exactly the experiments a confirmation-biased policy avoids. This bias arises in two equivalent ways: through the epistemic confirmation bias γ of Albarracin et al. (2022), and through a small affective utility $u(\theta)$ tilting the planning belief, so that discriminating experiments appear risky to the preferred paradigm and are under-sampled.

3.5 Trust and the social network

The agent is one of N on a graph $\mathcal{G} = (V, E)$, each carrying its own $(q_i(s_t \mid \theta), q_i(\theta), \lambda_i, U_i)$. Each agent emits its paradigm belief, and agent i forms a social observation as the trust-weighted mixture of its neighbours’ emissions, $o_i^s = \sum_j T_{ij} q_j(\theta)$, where T is the row-stochastic *trust matrix*, constructed from the graph adjacency A by row normalisation, $T = \text{rownorm}(A)$. The matrix carries no self term: an agent’s reliance on its own current belief is already represented by the prior in the slow update (7), weighted against incoming evidence by the conservatism λ , so a self-weight in T would duplicate that inertia. The trust matrix therefore encodes only which neighbours an agent attends to, a role distinct from λ ; the strength of the social channel is set by a message precision κ , and the confirmation bias γ up-weights aligned neighbours. The social signal enters the slow update (7) as a second accuracy channel. Figure 2 shows a Watts–Strogatz

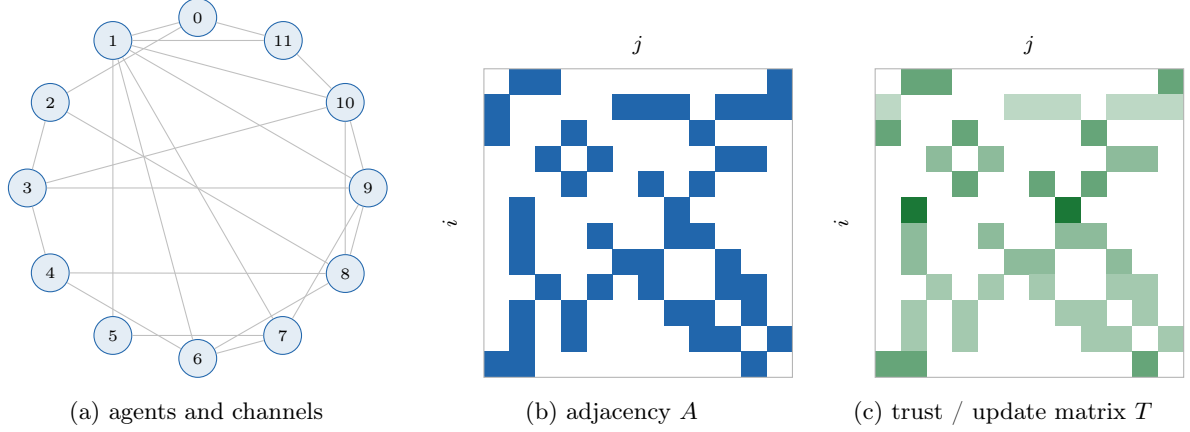


Figure 2: The trust network, from the Watts–Strogatz construction ($N=12$, $k=4$, rewiring 0.25) used in the implementation. (a) Agents (nodes) and their communication channels (edges). (b) The binary adjacency matrix A : a filled cell at (i, j) marks a channel. (c) The row-stochastic trust matrix $T = \text{rownorm}(A)$, whose row i gives the weights with which agent i mixes its neighbours’ paradigm beliefs into its social observation; darker cells carry more weight, and the diagonal is empty because own-belief inertia is supplied by the conservatism λ , not by T . Iterating the update propagates beliefs along these channels.

instance, its adjacency A , and the induced trust matrix T from the graph construction used in the implementation. The macroscopic object of interest is the *distribution* of the paradigm beliefs $\{q_i(\theta)\}$ across the population.

3.6 Regimes of slow paradigm change

The model makes the slowness of paradigm change a consequence of four distinct, separately measurable causes, three at the level of the individual and one social:

1. **Conservatism** λ large—the archive-recoding cost of Eq. (7) makes the paradigm update slow even when evidence is decisive.
2. **Structural ambiguity**—small marginal-likelihood ratios, i.e. low discriminability $d(a)$ (Eq. 3); the world does not separate the paradigms.
3. **Selective attention**—a policy biased away from discriminating experiments (Section 3.4), so the sharpening data are not gathered.
4. **Social inertia**—neighbours have not moved either, so the social channel pulls back toward the prevailing paradigm.

Each can produce the appearance of a stuck community without any individual being unreasonable. The extreme limits— $\lambda \rightarrow \infty$, or an affective utility that overpowers the accuracy term—yield permanent lock-in, but these are singular; for finite conservatism and adequate discriminating evidence the paradigm belief still moves, and the population converges. The phenomenon to be characterised is therefore not lock-in but the *transient*.

4 Anticipated results

The curves below are illustrative schematics, not simulation output; they specify the target phenomenology a numerical study of the model is intended to produce. The macroscopic variable is the population’s marginal paradigm belief. With one-round transmission delays through the graph (Section 3.5), the coupled population is a second-order system, and its approach to

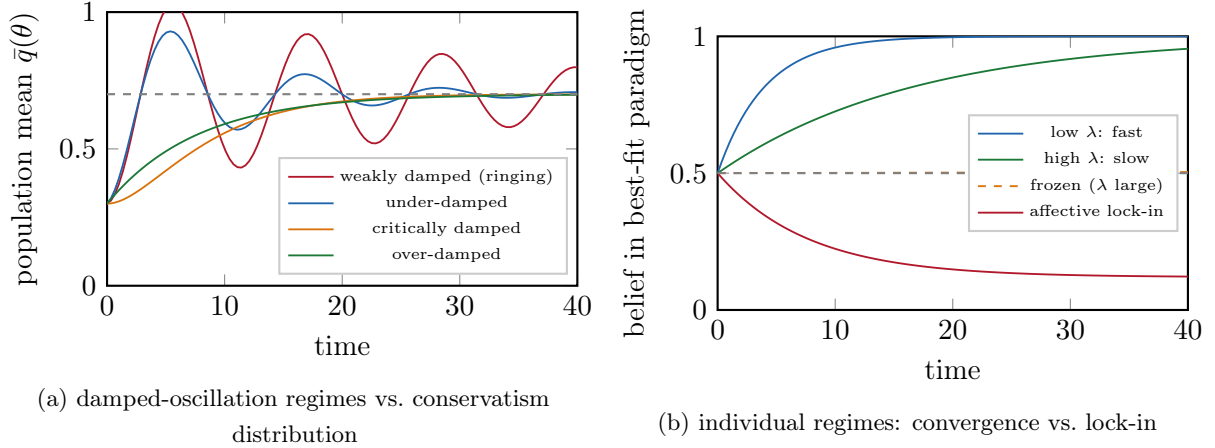


Figure 3: Illustrative target dynamics (schematic; not simulation output). (a) The population’s mean paradigm belief approaches consensus (dashed line) as a damped oscillation; the damping regime—weakly damped ringing, under-damped, critically damped, over-damped—is set by the distribution of conservatism $\{\lambda_i\}$ and the network structure. (b) At the individual level, finite conservatism yields convergence to the best-fit paradigm (fast or slow). All trajectories start from a neutral belief: under unbounded conservatism the belief stays frozen at its prior, while a supra-threshold affective utility pulls the belief toward the preferred non-best-fit paradigm, so belief in the best-fit paradigm declines toward a low fixed point.

consensus is a damped oscillation whose damping ratio is set by the distribution of conservatism $\{\lambda_i\}$ and by the spectral structure of T . Figure 3(a) shows the family of damping regimes the model should exhibit; panel (b) shows the individual-level belief trajectories distinguishing convergence from the two lock-in extremes.

5 Discussion

The two-timescale construction yields a layered answer to why communities do not switch paradigms when the evidence turns. The marginal likelihood favouring an alternative is often small because the world is structurally ambiguous; the policy does not seek the experiments that would sharpen it because attention is tilted by preference; even when discriminating evidence arrives, the cost of re-coding the accumulated record makes the paradigm update slow; and neighbours have not moved either. Four causes, four timescales, one model. None requires an agent to be irrational: each agent performs exact Bayesian inference over the world state and revises its theory at a rate its costs justify.

Two features distinguish the account. First, separating the fast world state from the slow paradigm locates theory-ladenness in a single interpretable place, the dependence of the likelihood on θ . Second, identifying the conservatism cost with the informational work of re-coding the archive gives a formal counterpart of incommensurability and grounds Hyland’s cost of mind-change in the retroactive reach of a paradigm over past data.

The model implies a concrete programme. The primary independent variable is the population distribution of conservatism $\{\lambda_i\}$; secondary variables are the experimental discriminability $d(a)$, the affective-utility distribution $\{U_i\}$, and the social coupling encoded in T , κ , and γ . The primary dependent variable is the damping of the population’s approach to consensus, with the time to consensus and the incidence of lock-in as derived quantities. The model predicts that lock-in does not occur for finite conservatism and sub-threshold affective utility regardless of social coupling, which is a falsifiable boundary, and that the qualitative dynamics are governed by the conservatism distribution rather than by any single agent’s parameters.

6 Conclusion

We have presented a categorical POMDP of paradigm persistence built on two latents at separate timescales: a fast world state with nontrivial dynamics and a slow, theory-laden paradigm maintained as a belief. World-state inference is exact, untempered Bayes; paradigm inference is a slow, motivated, tempered update whose conservatism cost is the work of re-coding the accumulated record. Networked agents propagate paradigm beliefs through a trust matrix, and the population’s distribution of paradigm beliefs approaches consensus as a damped oscillation set by the distribution of conservatism. The model converts the question of permanent capture into a question about the damping of convergence and identifies the distribution of resistance to changing one’s mind as the variable that controls it.

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